

JUVA PARTALA, Finland

WMO: 027360

Lat: 61.892N

Lon: 27.885E

Elev: 366

StdP: 14.50

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-10.6	-22.8	1.6	-17.1	-15.4	2.4	-10.5	15.4	20.9	13.8	22.1	1.5	260	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	81.2	66.6	77.6	64.9	74.4	62.7	69.1	77.8	67.0	74.7	64.8	72.1	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	97.1	72.6	63.9	90.3	71.4	61.7	83.3	69.0	33.4	77.8	31.7	74.4	30.0	72.6	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.7	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	14.9	20.7	26.1	37.2	50.3	57.3	62.4	59.4	50.7	38.5	31.7	23.3		
	DBStd	18.02	13.95	12.07	9.64	6.50	8.16	6.92	6.31	5.66	5.89	6.76	7.89	11.97		
	HDD50	4949	1088	821	740	384	100	15	4	62	358	549	827			
	HDD65	9393	1553	1241	1205	834	459	246	127	188	429	820	999	1292		
	CDD50	1110	0	0	0	1	108	233	387	296	82	3	0	0		
	CDD65	78	0	0	0	0	2	13	48	15	0	0	0	0		
	CDH74	780	0	0	0	0	60	158	432	130	0	0	0	0		
	CDH80	134	0	0	0	0	3	21	84	26	0	0	0	0		
	WSAvg	5.0	4.9	5.4	5.5	5.1	5.3	5.2	4.3	4.1	4.4	4.9	5.2	5.4		
	PrecAvg	24.3	1.9	1.5	1.3	1.2	1.6	2.6	3.1	2.7	2.0	2.4	2.2	1.9		
	PrecMax	30.4	3.3	3.2	2.5	2.6	4.8	4.4	4.8	5.5	4.8	5.7	4.6	3.5		
(15) Precipitation	PrecMin	19.6	0.3	0.2	0.2	0.4	0.4	0.8	1.0	0.6	0.7	0.5	0.4	0.7		
	PrecStd	2.8	0.7	0.7	0.6	0.6	1.0	0.8	1.1	1.3	1.1	1.1	0.9	0.7		
	DB	36.9	41.5	48.7	63.7	79.5	82.7	86.2	83.5	69.1	55.7	49.5	44.4			
	MCWB	35.3	37.4	38.4	49.1	60.2	65.1	69.8	68.9	60.7	52.4	47.1	42.4			
	2% DB	35.3	37.8	44.6	56.6	75.2	78.6	82.3	77.3	65.2	52.8	46.0	40.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	34.3	35.2	37.7	45.3	57.9	64.0	68.0	65.9	58.1	50.2	44.7	38.8			
	5% DB	33.2	35.7	40.4	52.4	71.1	74.8	79.1	73.6	62.6	51.1	43.4	37.4			
	MCWB	32.4	34.3	35.2	42.8	56.4	61.7	67.1	62.5	56.9	49.0	42.5	36.2			
	10% DB	32.1	34.0	37.6	48.7	66.4	71.2	75.3	70.3	60.0	48.7	41.1	35.3			
	MCWB	31.5	33.1	34.3	41.3	54.1	60.4	65.2	61.8	55.9	46.8	40.2	34.3			
	0.4% WB	36.0	37.8	40.6	50.1	63.9	69.3	72.5	70.7	61.7	53.3	47.6	43.0			
	MCDB	36.6	40.0	45.1	59.0	74.9	78.6	81.5	81.1	65.6	55.2	49.0	43.8			
	2% WB	34.4	35.8	38.0	46.9	60.8	66.0	70.2	67.2	59.6	51.3	44.6	38.8			
	MCDB	35.1	37.3	42.6	54.3	69.4	75.6	79.4	74.4	63.1	52.4	45.7	39.4			
	5% WB	32.8	34.4	36.3	44.3	58.0	63.4	68.3	64.8	58.0	49.5	42.6	36.3			
(33) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	33.4	35.5	39.6	50.3	67.1	72.2	76.3	71.5	61.2	50.6	43.5	37.1			
	10% WB	31.6	33.1	34.8	42.2	55.5	60.9	66.4	62.8	56.5	46.7	40.4	34.4			
	MCDB	32.2	34.0	37.0	47.6	65.8	69.2	73.5	68.4	59.4	48.5	41.0	35.1			
	5% DB	10.8	10.4	14.7	16.1	18.6	17.0	16.5	15.9	13.0	9.1	6.3	7.7			
	MCDBR	6.4	8.7	16.7	22.9	27.1	23.6	22.1	22.2	17.6	12.0	8.2	7.0			
(36) Mean Daily Temperature Range	MCWBR	6.4	7.4	11.2	13.6	13.0	11.4	10.5	11.5	11.3	9.5	8.0	7.1			
	5% WB	6.5	8.2	13.4	19.2	22.7	20.9	20.2	18.7	14.6	10.3	8.4	7.0			
	MCDBR	6.7	7.4	9.7	12.1	11.9	11.2	10.6	10.3	9.7	9.0	8.3	7.1			
	Clear-Sky Solar Irradiance	taub	0.242	0.290	0.317	0.356	0.353	0.350	0.374	0.358	0.335	0.317	0.263	0.359		
	taud	2.046	2.129	2.197	2.341	2.446	2.482	2.432	2.460	2.516	2.409	2.140	2.618			
(42) All-Sky Solar Radiation	Ebn at Noon	138	209	249	264	276	278	267	262	248	205	130	90			
	Edh at Noon	12	22	30	32	31	31	32	28	22	17	11	7			
	RadAvg	69	251	727	1167	1563	1671	1617	1243	711	286	75	34			
(45)	RadStd	11	60	96	130	177	103	159	149	82	47	13	4			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	-310	N/A	N/A	N/A

Nomenclature: See separate page